

How Search Engines Work

Three separate systems that fail in three different ways, and why knowing which one broke is most of the job.

Bottom line

Crawling, indexing and ranking are not three stages of one process. They are three independent systems, each with its own failure mode. A page can be crawled and never indexed, or indexed and never ranked, and those two problems share no fixes. Almost every SEO diagnosis starts by working out which of the three actually broke.

The pipeline, and the gates between the stages

Crawling is discovery. A bot finds a URL, requests it, and receives whatever the server returns. Googlebot finds URLs by following links, reading XML sitemaps, and taking manual submissions in Search Console.

Indexing is storage and understanding. Google processes the fetched page, works out what it is about, decides whether it is a duplicate of something it already has, and if it passes a quality bar, stores it. Google's index holds over **100 million gigabytes**.

Ranking is retrieval and ordering. When someone searches, Google pulls matching pages from the index and orders them using many separate systems.

The word to hold onto is **gate**. There is a gate between each stage, and each gate rejects things for its own reasons:

- Crawled does not mean indexed. Google routinely fetches a page, looks at it, and declines to store it.
- Indexed does not mean ranked. Being in the index just makes you eligible.
- Ranked does not mean visible. That is the whole AI-answer problem from Section 1.

Crawling: the technical limits that quietly bite

Most people picture crawling as a bot that reads your page like a person. It is closer to a program on a budget.

- **Googlebot fetches up to 2MB per URL.** PDFs get 64MB. Anything past that byte limit, including content and even HTML in the head, is simply not seen. A bloated page can have its real content sitting past the cutoff. [practitioner]

- **Crawling happens in two waves.** Wave one fetches the raw HTML. Wave two queues the page for the rendering service, which executes JavaScript. In 2026 the gap between them runs **24 to 72 hours**. [practitioner]
- **AI crawlers are different bots and often skip JavaScript entirely.** GPTBot, OAI-SearchBot, ClaudeBot and PerplexityBot crawl independently of Googlebot, and several of them do not execute JS at all. [practitioner]

That last point is the most under-appreciated fact in this whole course. If your content is rendered client-side, Google will eventually see it after wave two, and several AI engines may never see it. You can be perfectly visible in Google and structurally invisible to ChatGPT for a purely technical reason that has nothing to do with your content.

Crawl budget, the thing beginners worry about most, is the thing that matters least for them. It only becomes a real constraint above roughly **10,000 to 50,000 pages**. If your site is 40 pages, crawl budget is not your problem and time spent on it is wasted.

Indexing: the two rejections that look identical and are not

Search Console reports two exclusion states that beginners constantly confuse. They are completely different problems.

"Discovered - currently not indexed" means Google knows the URL exists but has not gotten around to fetching it. That is a **crawling** signal: budget, architecture, or internal linking. The page is buried or the site is too big for the attention it gets.

"Crawled - currently not indexed" means Google fetched the page, read it, and chose not to store it. That is a **quality** signal. Nothing technical is broken. Google looked at your page and decided it was not worth an index slot.

The fixes could not be more different. The first is solved with internal links and a flatter architecture. The second is solved by making the page better or deleting it. Getting this backwards is the single most common wasted week in SEO.

Indexing also does **canonicalization**: when Google finds several URLs with near-identical content, it picks one to represent the group. You express a preference with a canonical tag, but it is a hint and Google can and does overrule it. Search Console tells you the "Google-selected canonical", which is the answer that matters.

Ranking: not an algorithm, a committee

There is no "the algorithm". Google runs **15 or more documented ranking systems** at once. The ones worth knowing by name:

System	What it does
RankBrain	Interprets queries it has never seen before
BERT	Understands word context and how phrasing changes meaning
MUM	Works across languages and formats, including images and video
PageRank	The original link-authority system, still running
Passage Ranking	Ranks a specific section inside a long page, not just the page
Helpful Content system	Folded into core ranking in 2024. Demotes content built for search engines rather than people
Freshness systems	Surfaces newer content for time-sensitive queries
Reviews system	Judges the depth and expertise of review content

Two of these change how you should write. **Passage Ranking** means a single strong section inside a long page can rank on its own, so burying a good answer deep in a page is less fatal than it used to be. **The Helpful Content system** means writing that is transparently aimed at ranking rather than at a reader is now a demotion signal rather than a neutral one.

What Google confirms, and the long list it does not

Google publicly confirms only **seven** ranking factors: backlinks, content quality and E-E-A-T, HTTPS, page speed and Core Web Vitals, mobile-friendliness, freshness, and page experience.

Everything else you will read is inference. Some of it is good inference. Much of it is a tool vendor's correlation study, and a few things are actively contradicted by Google's own statements. These are **not** ranking factors, despite being repeated constantly:

- Bounce rate, as measured in Google Analytics
- Domain age
- Social signals such as likes and shares
- XML sitemaps as a ranking input, as opposed to a discovery aid
- Meta keywords
- Word count

That last one deserves a sentence, because it drives a lot of bad writing. Long pages correlate with better rankings. They correlate because thorough answers tend to be longer, not because length itself is rewarded. Writing to a word count produces padding, and padding is exactly what the Helpful

Content system is designed to catch.

The other engines

Google is not the only index. **Bing** powers ChatGPT's search, which makes it far more strategically important than its market share suggests. **IndexNow** is a push protocol supported by Bing, Yandex and Naver: instead of waiting to be crawled, you notify them that a URL changed. Google does not support it.

Why this matters

every diagnosis you will ever run starts with "which stage failed". Not crawled, crawled but not indexed, indexed but not ranked, and ranked but not clicked are four different problems with four different fixes. People who skip this section spend years applying ranking fixes to indexing problems and wondering why nothing works.

Do this now

- 1 Open Google Search Console for your capstone site.** If it is not verified, verify it now. Everything from here depends on it.
- 2 Run URL Inspection on your homepage.** Read the four things it tells you: whether the URL is on Google, the Google-selected canonical, the crawl date, and the page's indexing status.
- 3 Click "Test live URL", then "View tested page", then the screenshot and HTML tabs.** This is Googlebot's view. Compare it against what you see in your browser. If content is missing from the rendered HTML, you have found a rendering problem, which is Section 26.
- 4 Open the Pages report and find the exclusion reasons.** Note how many URLs sit in "Discovered - currently not indexed" versus "Crawled - currently not indexed". Write both numbers down.
- 5 Pick one page from "Crawled - currently not indexed" and read it as a stranger.** Be honest about why Google declined to store it. This is usually uncomfortable and usually correct.
- 6 Check `yoursite.com/robots.txt` in a browser.** Confirm it loads and returns content. You do not need to understand it yet, that is Section 21. Just confirm it exists and is not blocking everything.

Capstone step

You now have Search Console verified, a confirmed index status for the homepage, both exclusion counts written down, and one honest assessment of a page Google refused to index. If your site turned out to have indexing problems, note them. Tier 3 is where you fix them, and you will want the before numbers.

Key takeaways

- Crawling, indexing and ranking are three separate systems with three separate failure modes. Diagnosis means identifying which one broke before touching anything.
- "Discovered - not indexed" is a crawling and architecture problem. "Crawled - not indexed" is a content quality problem. They look similar in Search Console and share no fixes.
- There is no single algorithm. At least fifteen documented ranking systems run at once, and two of them, Passage Ranking and the Helpful Content system, should change how you write.
- Google confirms only seven ranking factors. Bounce rate, domain age, social signals and word count are not among them, regardless of how often you read otherwise.